20th European Control Conference (ECC),
July 12-15 2022, London, UK

A Multirate Variational Approach to Nonlinear MPC

Yana Lishkova , DPhil Candidate*
Prof. Mark Cannon*
Prof. Sina Ober-Blöbaum §

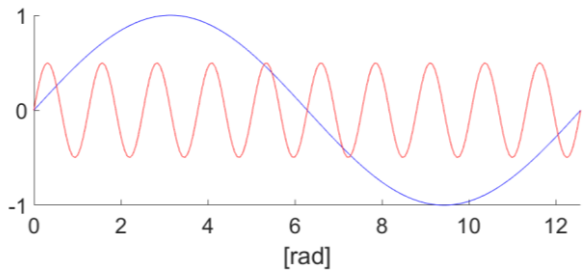
* Department of Engineering Science, University of Oxford
§ Department of Mathematics, University of Paderborn



Outline

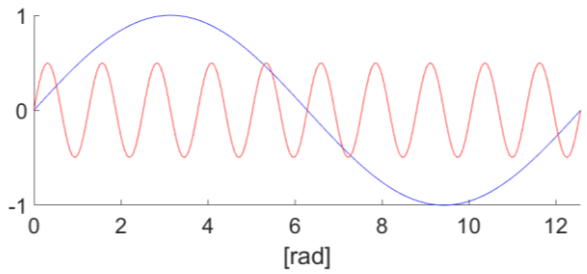
- Overview
- Multirate systems and models
- Variational integrators
- Multirate variational NMPC
- Case studies and discussion

MPC for Multirate systems

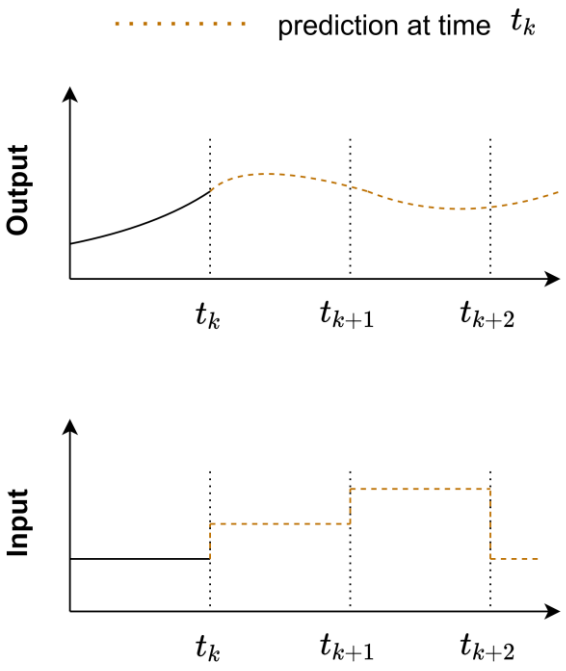


- Multirate systems exhibit dynamics on different time scales

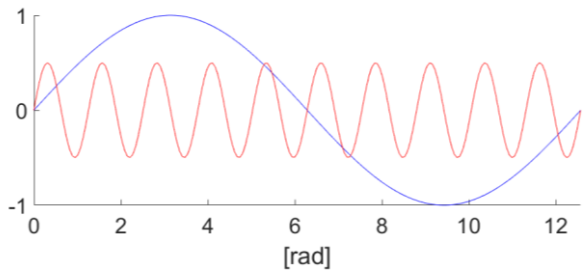
MPC for Multirate systems



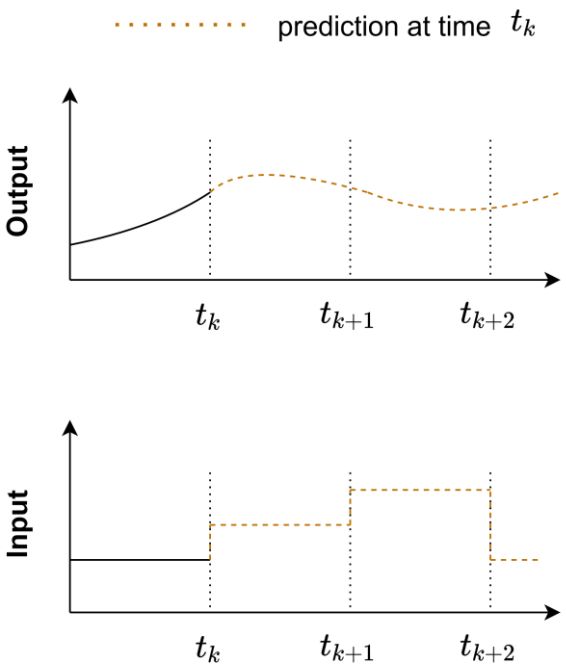
- Multirate systems exhibit dynamics on different time scales
- MPC- Model predictive control / Receding horizon control

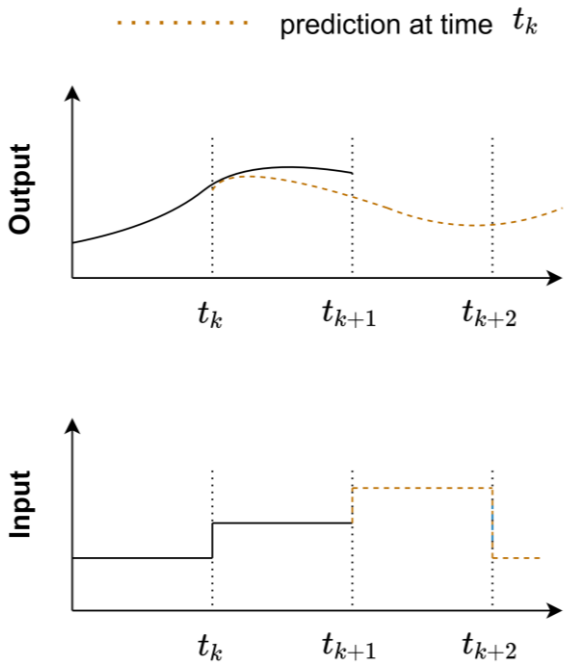


MPC for Multirate systems

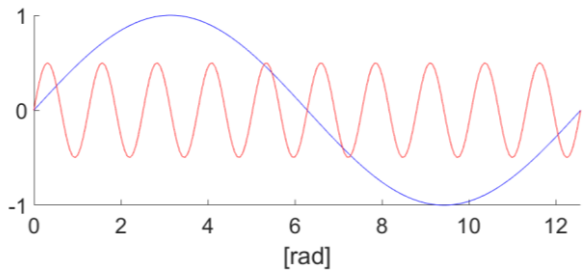


- Multirate systems exhibit dynamics on different time scales
- MPC- Model predictive control / Receding horizon control

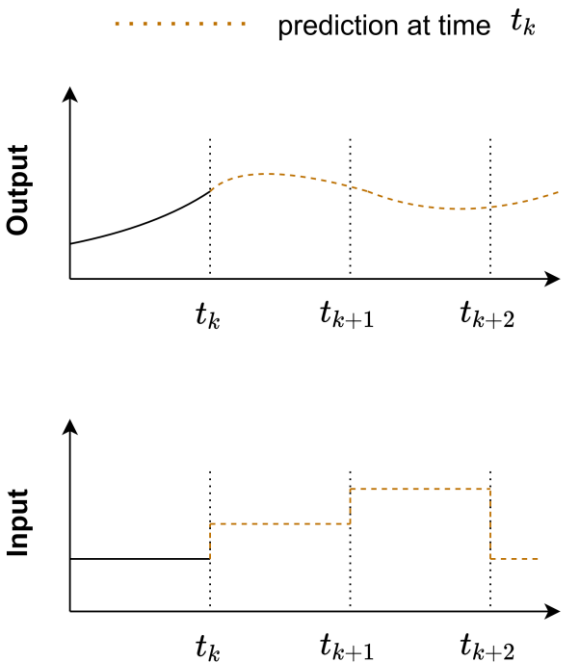


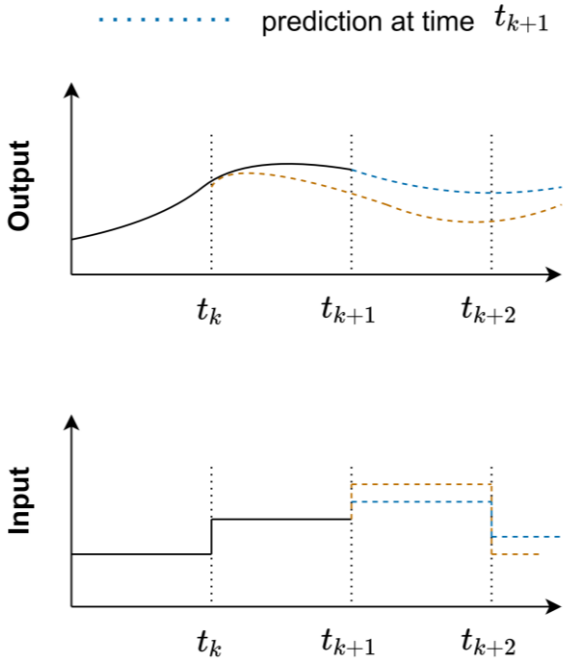


MPC for Multirate systems

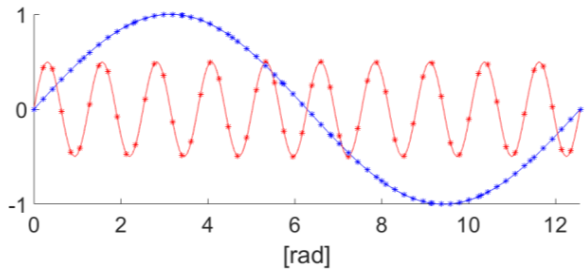


- Multirate systems exhibit dynamics on different time scales
- MPC- Model predictive control / Receding horizon control

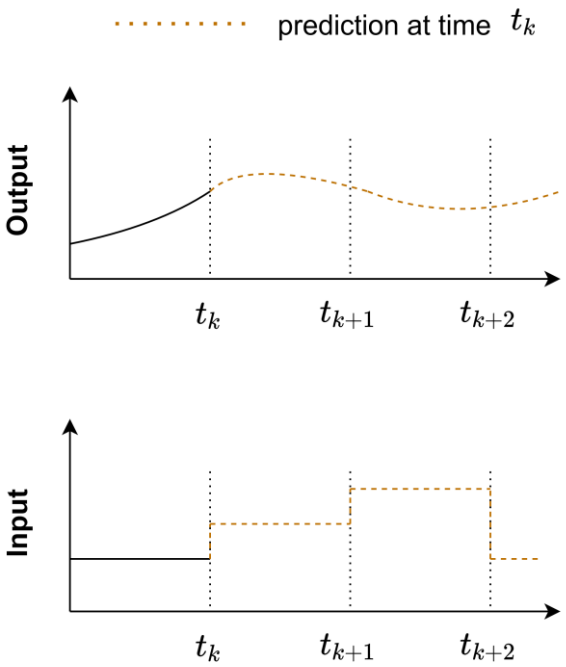


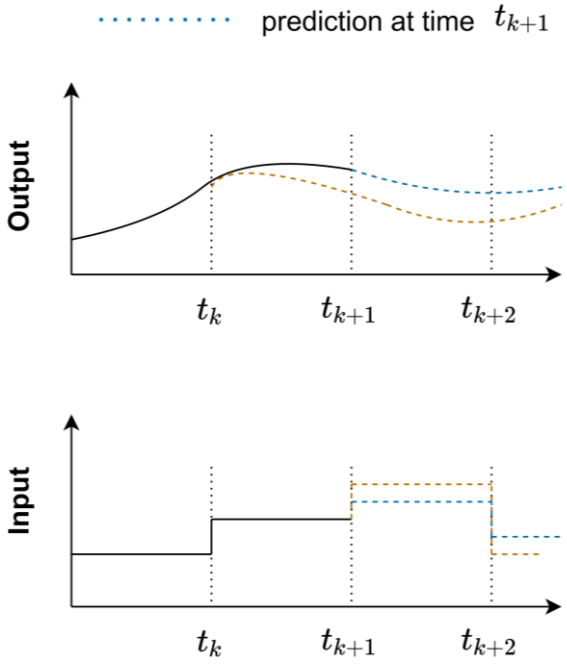


MPC for Multirate systems

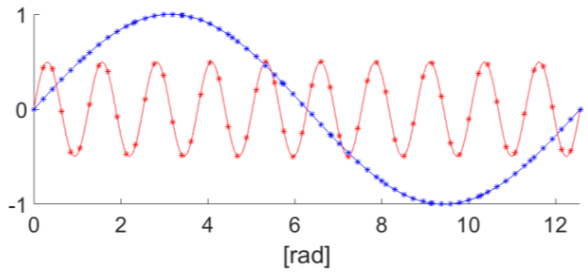


- Multirate systems exhibit dynamics on different time scales
- MPC- Model predictive control / Receding horizon control





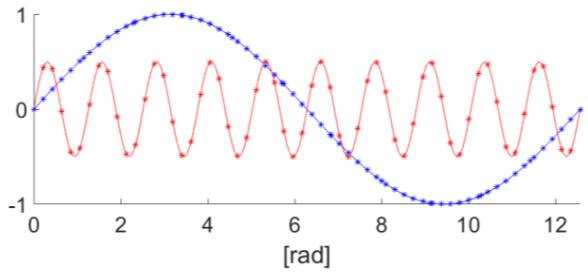
MPC for Multirate systems



- Multirate systems exhibit dynamics on different time scales
- MPC- Model predictive control / Receding horizon control

- ***Available approaches:***
  - Non-uniform discretization schemes
  - Move-blocking strategies
  - And others

MPC for Multirate systems

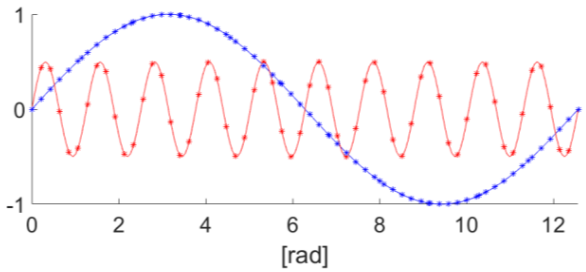


- Multirate systems exhibit dynamics on different time scales
- MPC- Model predictive control / Receding horizon control

- ***Available approaches:***
  - Non-uniform discretization schemes
  - Move-blocking strategies
  - And others

- ***We propose the use of multirate variational MPC formulation:***

MPC for Multirate systems



- Multirate systems exhibit dynamics on different time scales
- MPC- Model predictive control / Receding horizon control

- ***Available approaches:***
  - Non-uniform discretization schemes
  - Move-blocking strategies
  - And others

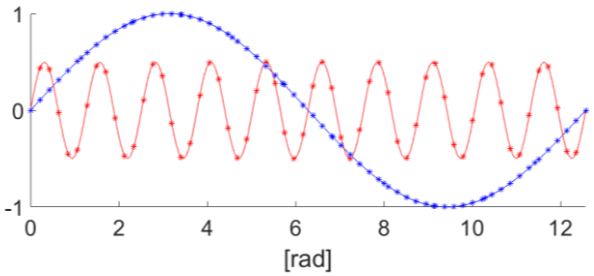
- ***We propose the use of multirate variational MPC formulation:***
  - Reduces the size of the OCP solved at each step
  - Allows for the OCP to be solved more rarely
  - Allows for high fidelity resolution of both slow and fast dynamics for exponentially long times

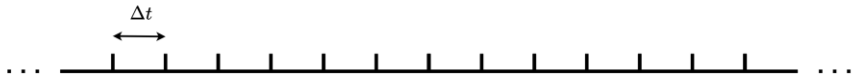
Multirate variational MPC

- **Centers on the use of multirate variational description of the system**

Multirate variational MPC

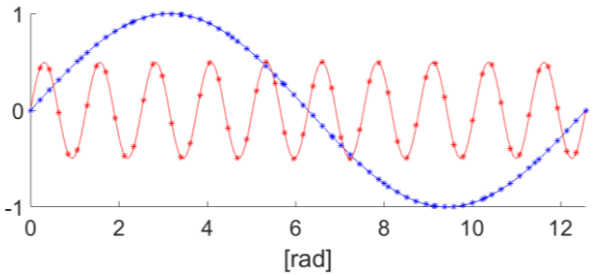
- **Centers on the use of multirate variational description of the system**

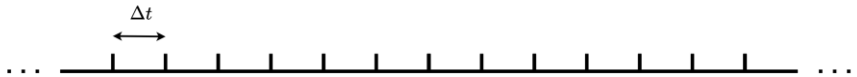




Multirate variational MPC

- **Centers on the use of multirate variational description of the system**



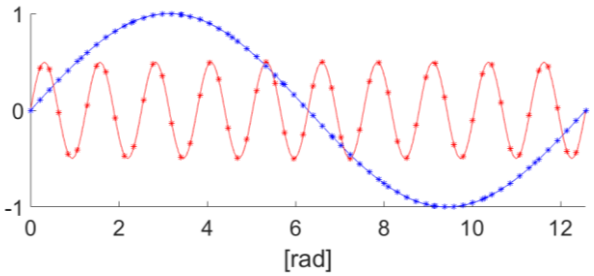


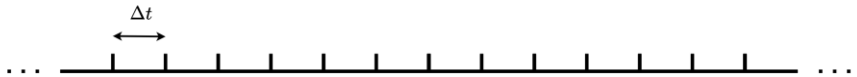
system dynamics

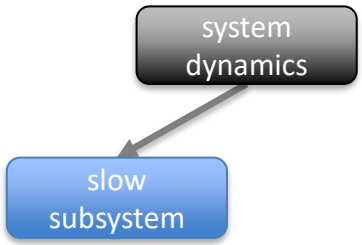
* **S. Leyendecker and S. Ober-Blöbaum**, "A Variational Approach to Multirate Integration for Constrained Systems," *Multibody Dynamics: Computational Methods and Applications* (J.-C. Samin and P. Fisette, eds.), pp. 97–121, Dordrecht: Springer Netherlands, 2013

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**



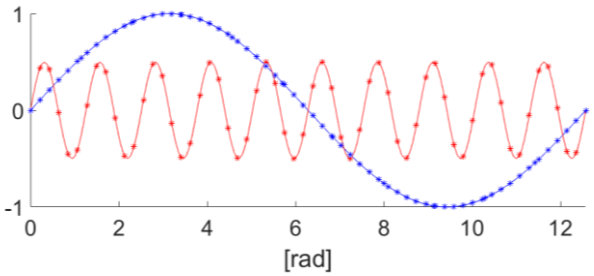


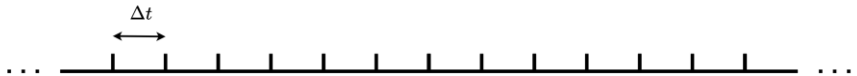


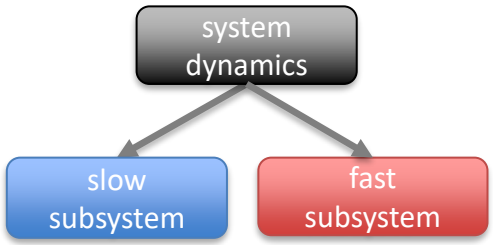
* **S. Leyendecker and S. Ober-Blöbaum**, "A Variational Approach to Multirate Integration for Constrained Systems," *Multibody Dynamics: Computational Methods and Applications* (J.-C. Samin and P. Fisette, eds.), pp. 97–121, Dordrecht: Springer Netherlands, 2013

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**



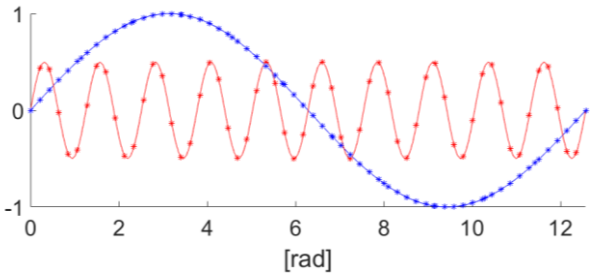


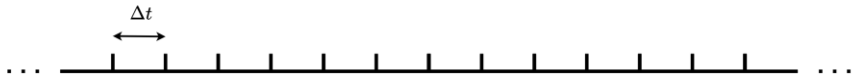


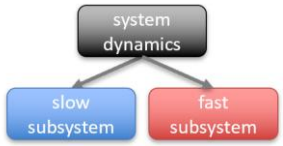
* **S. Leyendecker and S. Ober-Blöbaum**, "A Variational Approach to Multirate Integration for Constrained Systems," *Multibody Dynamics: Computational Methods and Applications* (J.-C. Samin and P. Fisette, eds.), pp. 97–121, Dordrecht: Springer Netherlands, 2013

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**

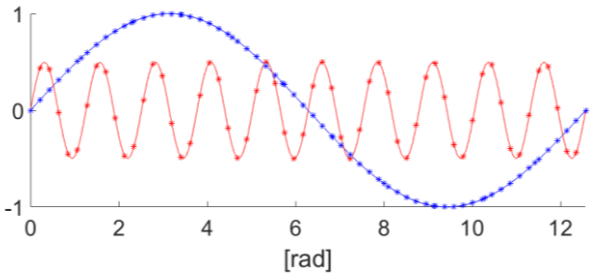


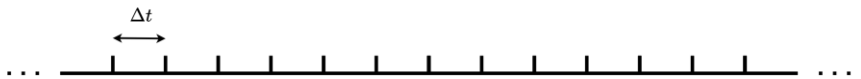


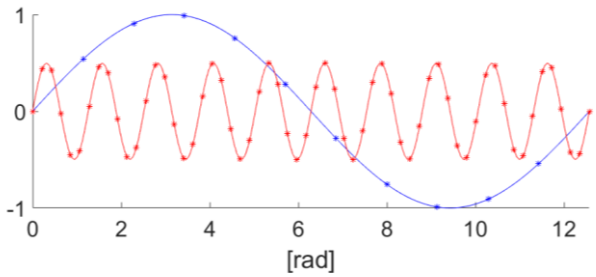


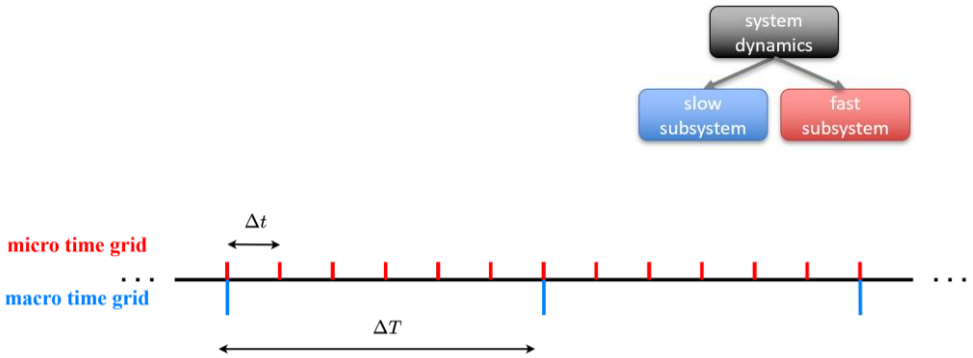
Multirate variational MPC

- **Centers on the use of multirate variational description of the system**









Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ **Multirate formulation** -> ***reduced OCP size, increased receding horizon time step***

    -> maintains recursive feasibility with high fidelity of the simulation

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ Multirate formulation -> *reduced OCP size, increased receding horizon time step*

    -> maintains recursive feasibility with high fidelity of the simulation

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ Multirate formulation -> *reduced OCP size, increased receding horizon time step*

    -> maintains recursive feasibility with high fidelity of the simulation
  - ✓ ***Variational approach***

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ Multirate formulation -> *reduced OCP size, increased receding horizon time step*

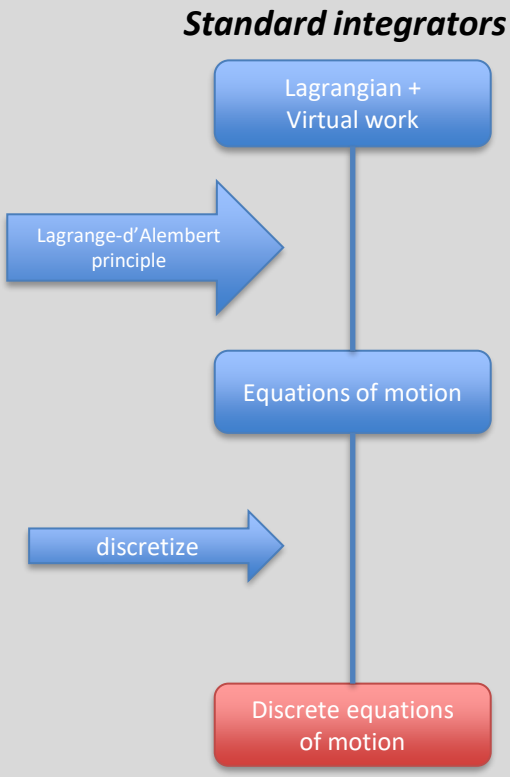
    -> maintains recursive feasibility with high fidelity of the simulation
  - ✓ **Variational approach**

    -> improved qualitative representation*

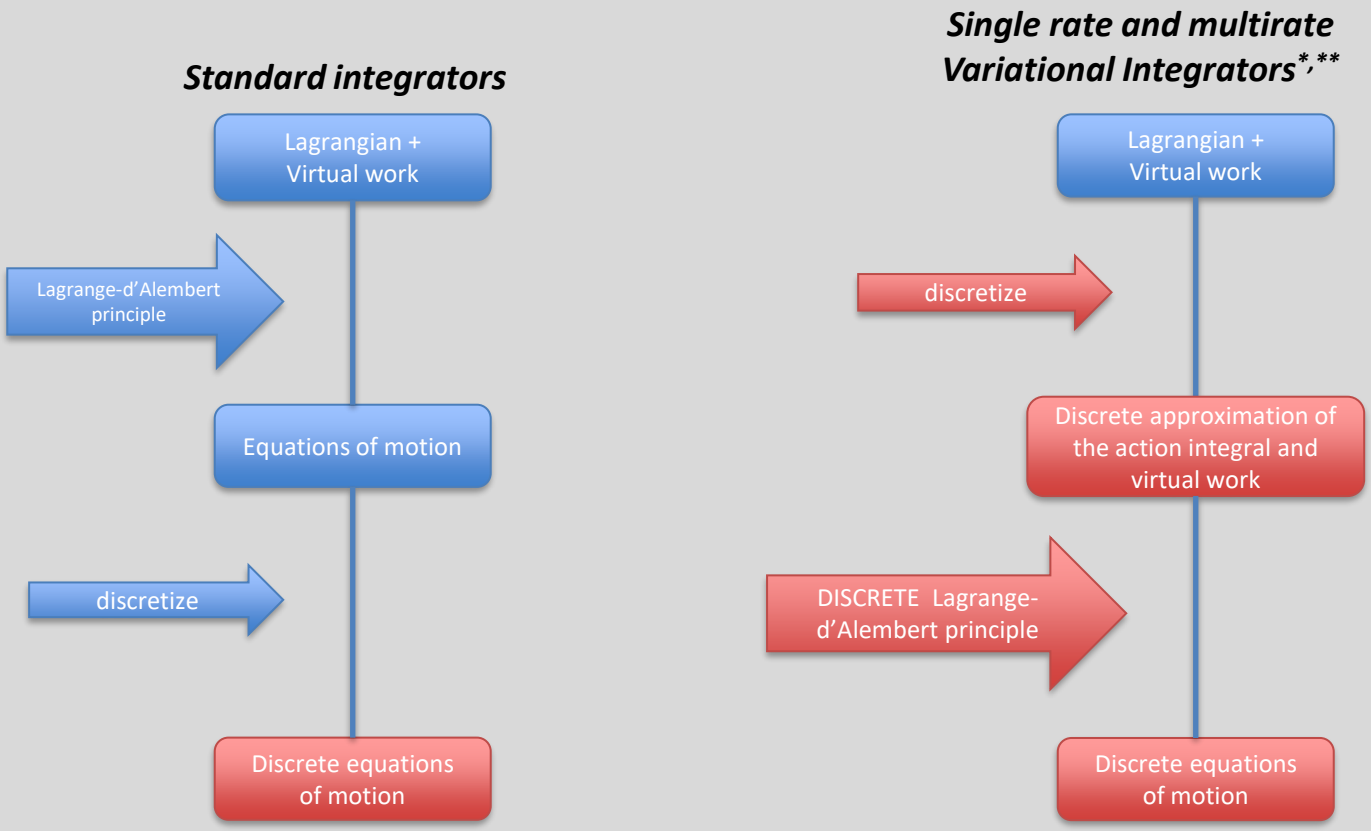
    -> accurate long-term integration

***Arieh Iserles.** *A First Course in the Numerical Analysis of Differential Equations.* Cambridge Texts in Applied Mathematics. Cambridge University Press, 2 edition, 2008.
****Dina Razandralandy, Aziz Hamdouni, and Marx Chhay.** A review of some geometric integrators. *Advanced Modeling and Simulation in Engineering Sciences*, 5(1):1-67, 2018.

Formulation comparison



Variational formulation



* **T. Gail, S. Ober-Blöbaum, and S. Leyendecker**, "Variational multirate integration in discrete mechanics and optimal control," *Proceedings of* ECCOMAS, 2017, pp. 1–4.
** **S. Ober-Blöbaum, O. Junge, and J. E. Marsden**, "Discrete mechanics and optimal control: an analysis," *ESAIM: Control, Optimisation and Calculus of Variations*, Vol. 17, No. 2, 2011, pp. 322–352.

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ Multirate formulation *-> reduced OCP size, increased receding horizon time step*

    -> maintains recursive feasibility with high fidelity of the simulation
  - ✓ **Variational approach** ***-> conservation of energy and/or momenta of the system***

    -> improved long-term accuracy

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ Multirate formulation -> *reduced OCP size, increased receding horizon time step*

    -> *maintains recursive feasibility with high fidelity of the simulation*
  - ✓ Variational approach -> *conservation of energy and/or momenta of the system*

    -> *improved long-term accuracy*

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ Multirate formulation -> *reduced OCP size, increased receding horizon time step*

    -> *maintains recursive feasibility with high fidelity of the simulation*
  - ✓ Variational approach -> *conservation of energy and/or momenta of the system*

    -> *improved long-term accuracy*
  - ✓ **No need for decoupling of the equations of motion -> avoidance of approximations, *more general applicability to a variety of linear and nonlinear models***
  - ✓ ***Provides sparse structure for the constraint Jacobian***

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ Multirate formulation -> *reduced OCP size, increased receding horizon time step*

    -> *maintains recursive feasibility with high fidelity of the simulation*
  - ✓ Variational approach -> *conservation of energy and/or momenta of the system*

    -> *improved long-term accuracy*
  - ✓ No need for decoupling of the equations of motion -> avoidance of approximations, *more general applicability to a variety of linear and nonlinear models*
  - ✓ *Provides sparse structure for the constraint Jacobian*

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ Multirate formulation -> *reduced OCP size, increased receding horizon time step*

    -> *maintains recursive feasibility with high fidelity of the simulation*
  - ✓ Variational approach -> *conservation of energy and/or momenta of the system*

    -> *improved long-term accuracy*
  - ✓ No need for decoupling of the equations of motion -> avoidance of approximations, *more general applicability to a variety of linear and nonlinear models*
  - ✓ *Provides sparse structure for the constraint Jacobian*
  - – ***For nonlinear dynamics the resulting models are nonlinear leading to nonlinear optimization problems***

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ Multirate formulation -> *reduced OCP size, increased receding horizon time step*

    -> *maintains recursive feasibility with high fidelity of the simulation*
  - ✓ Variational approach -> *conservation of energy and/or momenta of the system*

    -> *improved long-term accuracy*
  - ✓ No need for decoupling of the equations of motion -> avoidance of approximations, *more general applicability to a variety of linear and nonlinear models*
  - ✓ *Provides sparse structure for the constraint Jacobian*
  - – *For nonlinear dynamics the resulting models are nonlinear leading to nonlinear optimization problems*

Multirate variational MPC

- **Centers on the use of multirate variational description of the system**
  - ✓ Multirate formulation -> *reduced OCP size, increased receding horizon time step*

    -> *maintains recursive feasibility with high fidelity of the simulation*
  - ✓ Variational approach -> *conservation of energy and/or momenta of the system*

    -> *improved long-term accuracy*
  - ✓ No need for decoupling of the equations of motion -> avoidance of approximations, *more general applicability to a variety of linear and nonlinear models*
  - ✓ *Provides sparse structure for the constraint Jacobian*
  - – *For nonlinear dynamics the resulting models are nonlinear leading to nonlinear optimization problems*

Novel multirate variational linearization method which maintains all the advantages of the full nonlinear representation listed above

Multirate variational MPC

- **Multirate variational reformulation of tube-based MPC**
  - **Tube-based approach- based on Cannon et al.[*]**

✓ Successive variation linearization of the system -> replaces the NLP at each discrete time step with an iteration of convex OCP formulation

✓ Robustly accounts for the error of this approximation - > *guarantees closed-loop stability of the scheme*

✓ *The tube can be optimized online reducing the conservativeness of the approach (Algorithm 2) or kept constant to further reduce the computational load of the scheme (Algorithm 1)*

[*] M. Cannon, J. Buerger, B. Kouvaritakis, and S. Rakovic, "Robust Tubes in Nonlinear Model Predictive Control," *IEEE Transactions on Automatic Control*, vol. 56, no. 8, pp. 1942–1947, 2011.

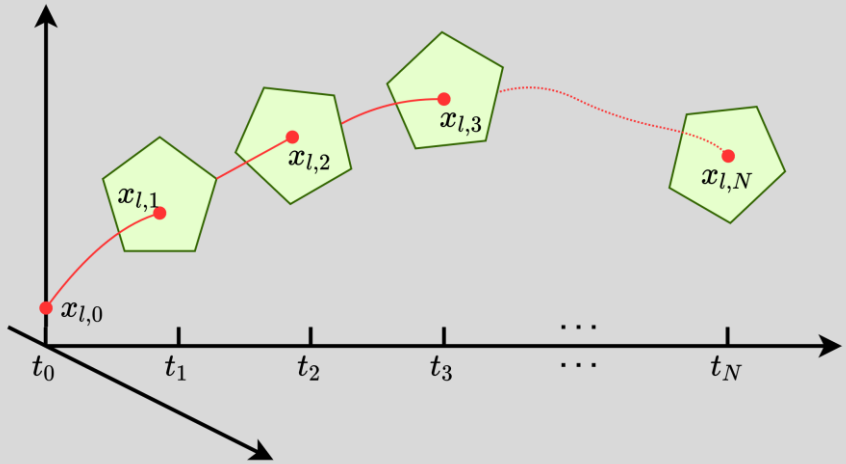
Multirate variational MPC

- **Multirate variational reformulation of tube-based MPC**
  - **Tube-based approach- based on Cannon et al.***

✓ Successive variation linearization of the system -> replaces the NLP at each discrete time step with an iteration of convex OCP formulation

✓ Robustly accounts for the error of this approximation - > *guarantees closed-loop stability of the scheme*

✓ *The tube can be optimized online reducing the conservativeness of the approach (Algorithm 2) or kept constant to further reduce the computational load of the scheme (Algorithm 1)*



* M. Cannon, J. Buerger, B. Kouvaritakis, and S. Rakovic, "Robust Tubes in Nonlinear Model Predictive Control," *IEEE Transactions on Automatic Control*, vol. 56, no. 8, pp. 1942–1947, 2011.

Multirate variational MPC

- **Tube-based approach**
  - ✓ Successive variation linearization of the system -> replaces the NLP at each discrete time step with an iteration of convex OCP formulation
  - ✓ Robustly accounts for the error of this approximation - *> guarantees closed-loop stability of the scheme*
  - ✓ *The tube can be optimized online reducing the conservativeness of the approach (Algorithm 2) or kept constant to further reduce the computational load of the approach (Algorithm 1)*

Multirate variational MPC

- **Tube-based approach**
  - ✓ Successive variation linearization of the system -> replaces the NLP at each discrete time step with an iteration of convex OCP formulation
  - ✓ Robustly accounts for the error of this approximation - *> guarantees closed-loop stability of the scheme*
  - ✓ *The tube can be optimized online reducing the conservativeness of the approach (Algorithm 2) or kept constant to further reduce the computational load of the approach (Algorithm 1)*

- **Multirate variational reformulation**

Multirate variational MPC

- **Tube-based approach**
  - ✓ Successive variation linearization of the system -> replaces the NLP at each discrete time step with an iteration of convex OCP formulation
  - ✓ Robustly accounts for the error of this approximation - *> guarantees closed-loop stability of the scheme*
  - ✓ *The tube can be optimized online reducing the conservativeness of the approach (Algorithm 2) or kept constant to further reduce the computational load of the approach (Algorithm 1)*

- **Multirate variational reformulation**
  - ✓ *Multirate formulation -> reduced OCP size, increased receding horizon time step*

    -> maintains recursive feasibility with high fidelity of the simulation

Multirate variational MPC

UNIVERSITY OF OXFORD

- **Tube-based approach**
  - ✓ Successive variation linearization of the system -> replaces the NLP at each discrete time step with an iteration of convex OCP formulation
  - ✓ Robustly accounts for the error of this approximation - *> guarantees closed-loop stability of the scheme*
  - ✓ *The tube can be optimized online reducing the conservativeness of the approach (Algorithm 2) or kept constant to further reduce the computational load of the approach (Algorithm 1)*

- **Multirate variational reformulation**
  - ✓ *Multirate formulation -> reduced OCP size, increased receding horizon time step*

    -> maintains recursive feasibility with high fidelity of the simulation
  - ✓ Variational linearization approach *-> conservation of energy and/or momenta of the system*

    -> improved long-term accuracy

    -> alternative means for bounding the error of this approximation

Multirate variational MPC

- **Tube-based approach**
  - ✓ Successive variation linearization of the system -> replaces the NLP at each discrete time step with an iteration of convex OCP formulation
  - ✓ Robustly accounts for the error of this approximation - *> guarantees closed-loop stability of the scheme*
  - ✓ *The tube can be optimized online reducing the conservativeness of the approach (Algorithm 2) or kept constant to further reduce the computational load of the approach (Algorithm 1)*

- **Multirate variational reformulation**
  - ✓ *Multirate formulation -> reduced OCP size, increased receding horizon time step*

    -> maintains recursive feasibility with high fidelity of the simulation
  - ✓ Variational linearization approach *-> conservation of energy and/or momenta of the system*

    -> improved long-term accuracy

    -> alternative means for bounding the error of this approximation
  - ✓ Unified control methodology without the need to decouple the equations of motion -> *potential improvement in optimality and constraint- handling capabilities*

Case studies

- **Planar quadcopter model**

$$\mathcal{L}(q,\dot{q}) = \tfrac{1}{2}\dot{q}^\top\dot{q} - [0,g,0]^\top \cdot q$$

$$f = [(u_1+g)\sin\alpha, (u_1+g)\cos\alpha, u_2]^\top$$

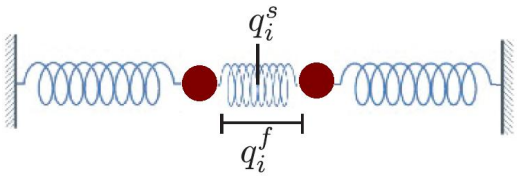
$$q = [y,z,\alpha]^\top \qquad u = [u_1,u_2]^\top$$

y, z, α - the horizontal, vertical and angular displacement

u_1, u_2 - controls proportional to the net thrust and moment acting on the quadcopter

- **The Fermi-Pasta-Ulam problem (FPU)**

$$\mathcal{L}(q^s,q^f,\dot{q}^s,\dot{q}^f) = \frac{(\dot{q}^s)^2+(\dot{q}^f)^2-(\eta q^f)^2}{2} - \frac{(q^s+q^f)^4+(q^s-q^f)^4}{4}$$



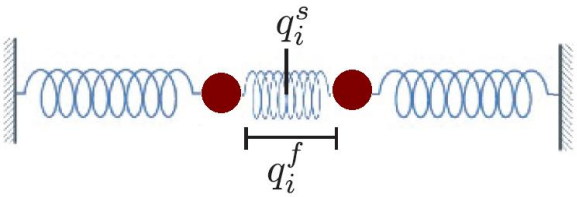
$$f^s = u^s$$
$$f^f = u^f$$

q^f, q^s - the length of the stiff spring and the location of its center

u^s, u^f - the resolution of the control applied to the *i*-th mass along q^s and q^f respectively

Case study

- **The Fermi-Pasta-Ulam problem**



$$q_0^s = \dot{q}_0^s = \dot{q}_0^f = 1 \text{ and } q_0^f = \tfrac{1}{\eta}$$

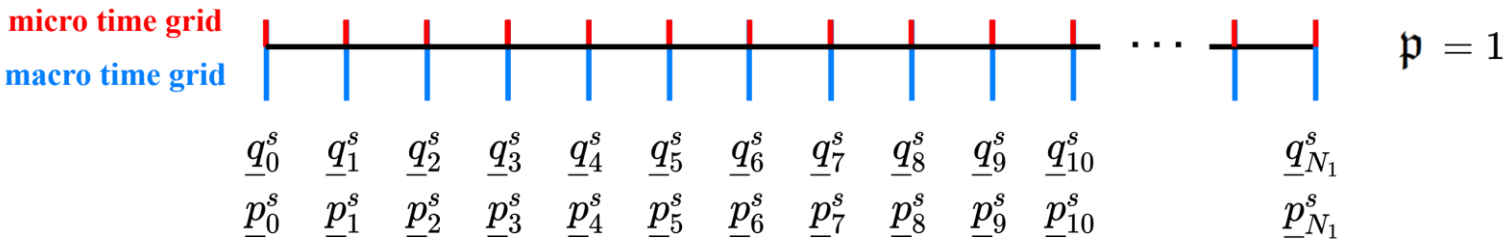
$$Q^M = \Delta T \operatorname{diag}\{1,1,1,1\} \qquad E[q^s, q^f]^\top \leq [10,10]^\top$$

$$R^M = \operatorname{diag}\{\Delta T, \Delta t, \Delta t, \ldots, \Delta t\} \qquad E[\dot{q}^s, \dot{q}^f]^\top \leq [10,10]^\top$$

$$E = [1/\sqrt{2}, -1/\sqrt{2}; 1/\sqrt{2}, 1/\sqrt{2}] \qquad E[u^s, u^f]^\top \leq [20,20]^\top$$

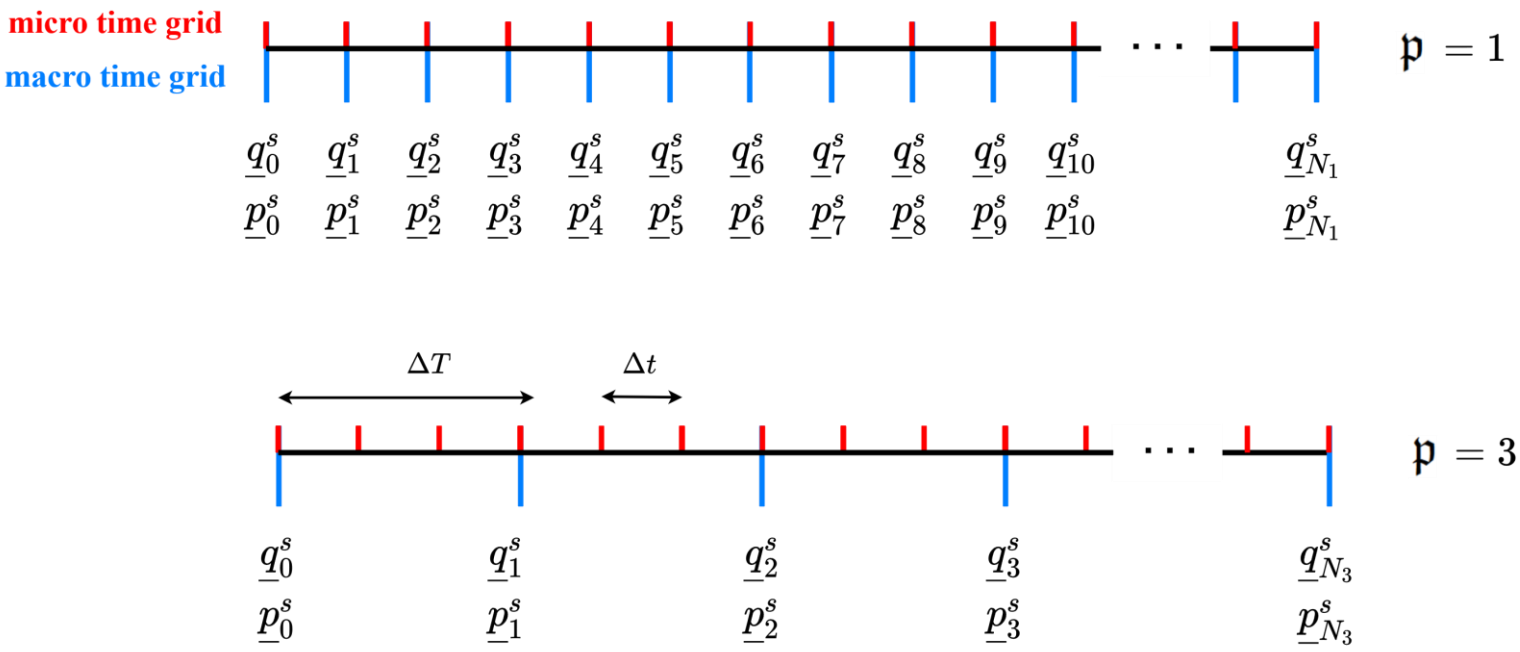
Increasing $\mathfrak{p}$

$$\underline{p} = \frac{\mathcal{L}}{\dot{\underline{q}}}, \quad \mathfrak{p} = \frac{\text{macro time step}}{\text{micro time step}} = \frac{\Delta T}{\Delta t}$$



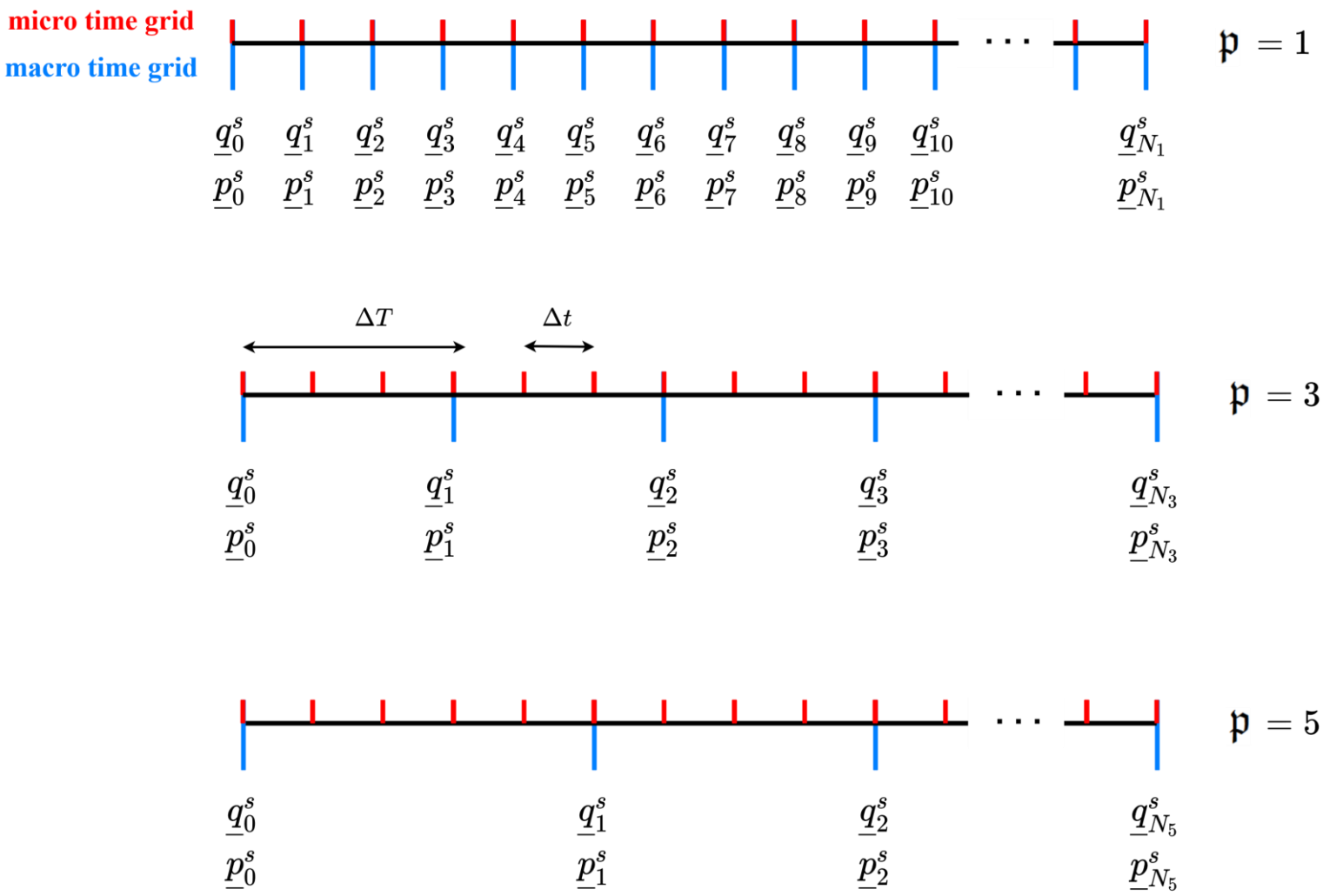
Increasing $\mathfrak{p}$

$$\underline{p} = \frac{\mathcal{L}}{\dot{\underline{q}}}, \quad \mathfrak{p} = \frac{\text{macro time step}}{\text{micro time step}} = \frac{\Delta T}{\Delta t}$$



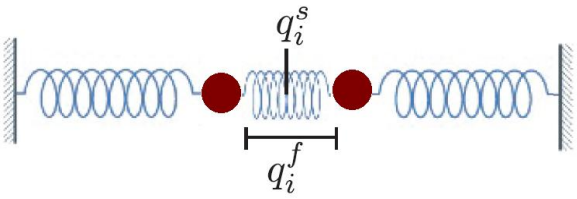
Increasing $\mathfrak{p}$

$$\underline{p} = \frac{\mathcal{L}}{\dot{\underline{q}}}, \quad \mathfrak{p} = \frac{\text{macro time step}}{\text{micro time step}} = \frac{\Delta T}{\Delta t}$$



Case study

The Fermi-Pasta-Ulam problem



$$q_0^s = \dot{q}_0^s = \dot{q}_0^f = 1 \text{ and } q_0^f = \tfrac{1}{\eta}$$

$$Q^M = \Delta T \,\mathrm{diag}\{1,1,1,1\} \qquad E[q^s, q^f]^\top \leq [10,10]^\top$$

$$R^M = \mathrm{diag}\{\Delta T, \Delta t, \Delta t, \ldots, \Delta t\} \qquad E[\dot{q}^s, \dot{q}^f]^\top \leq [10,10]^\top$$

$$E = [1/\sqrt{2}, -1/\sqrt{2}; 1/\sqrt{2}, 1/\sqrt{2}] \qquad E[u^s, u^f]^\top \leq [20,20]^\top$$

Simulations for $\Delta t = 10^{-3}$, $t_f = 0.01$, $t_{target} = 30\Delta t$ lead to the following results:

n_{rhocp} - number of discrete time steps until reaching t_{target}

ONLINE COMPUTATIONAL LOAD

	Algorithm 1			Algorithm 2	
$\mathfrak{p}$	1	3	5	1	3
CPU time per iter at $i = 0$	**3.18** ± 0.03	**2.57** ± 0.48	**1.49** ± 0.02	**5.52** ± 0.17	**4.46** ± 0.01
n_{rhocp}	30	10	6	30	10
CPU time to reach t_{target}	**96.3** ± 0.24	**27.47** ± 0.43	**11.14** ± 0.89	**173.79** ± 1.59	**51.17** ± 0.08

Comparable simulations for a single-rate and multirate variational nonlinear programming solution solution are $43.57, 31.76, 29.55$ for $\mathfrak{p} = 1, 3, 5$ respectively.

Multirate variational MPC (Conclusions)

- **Tube-based approach**
  - ✓ Successive variation linearization of the system -> replaces the NLP at each discrete time step with an iteration of convex OCP formulation
  - ✓ Robustly accounts for the error of this approximation - *> guarantees closed-loop stability of the scheme*
  - ✓ *The tube can be optimized online reducing the conservativeness of the approach (Algorithm 2) or kept constant to further reduce the computational load of the approach (Algorithm 1)*

- **Multirate variational reformulation**
  - ✓ *Multirate formulation -> reduced OCP size, increased receding horizon time step*

    -> maintains recursive feasibility with high fidelity of the simulation
  - ✓ Variational linearization approach *-> conservation of energy and/or momenta of the system*

    -> improved long-term accuracy

    -> alternative means for bounding the error of this approximation
  - ✓ Unified control methodology without the need to decouple the equations of motion -> *potential improvement in optimality and constraint- handling capabilities*

THANK YOU FOR YOUR ATTENTION!

Contact details:

Yana Lishkova, yana.lishkova@eng.ox.ac.uk

Address: St. Hugh's College, St Margaret's Road, OX2 6LE, United Kingdom

Main references:

[1] T. Gail, S. Ober-Blöbaum, and S. Leyendecker, "Variational multirate integration in discrete mechanics and optimal control," *Proceedings of* ECCOMAS, 2017, pp. 1–4.

[2] M. Cannon, J. Buerger, B. Kouvaritakis, and S. Rakovic, "Robust Tubes in Nonlinear Model Predictive Control," *IEEE Transactions on Automatic Control*, vol. 56, no. 8, pp. 1942–1947, 2011.

[3] O. Junge, J. E. Marsden, and S. Ober-Blöbaum, "Discrete mechanics and optimal control," IFAC *Proceedings Volumes*, Vol. 38, No. 1, 2005, pp. 538–543.

[4] S. Ober-Blöbaum, O. Junge, and J. E. Marsden, "Discrete mechanics and optimal control: an analysis," *ESAIM: Control, Optimisation and Calculus of Variations*, Vol. 17, No. 2, 2011, pp. 322–352.

[5] S. Leyendecker and S. Ober-Blöbaum, "A Variational Approach to Multirate Integration for Constrained Systems," *Multibody Dynamics: Computational Methods and Applications* (J.-C. Samin and P. Fisette, eds.), pp. 97–121, Dordrecht: Springer Netherlands, 2013

[6] T. Gail, S. Leyendecker, and S. Ober-Blöbaum, "Computing time investigations of variational multirate integrators," *ECCOMAS Multibody Dynamics*, 2013.

[7] Y. Lishkova, S. Ober-Blöbaum, M. Cannon, and S. Leyendecker**, "**A multirate variational approach to simulation and optimal control for flexible spacecraft*", in AAS Advances in the Astronautical Sciences, Volume 175, Astrodynamics 2020*